41 An attempt at a hypothetical 3D reconstruction of one of the 13 Djeddar Funerary Mausoleums in Tiaret, Algeria Using Aerial Photogrammetry.

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The archaeological site of Jedar, which is located in the province of Tiaret. South-west of the Algerian capital, consists of 13 funerary mausoleums, three of which are in Al-Akhdar mountains and the remaining ten in Laroui mountains, where the largest of them is located, and only four of them preserve their architectural remain, the geometric composition of them consists of a square base and a stepped pyramid, while their interior design varies from one mausoleum to another, the first consists of 8 funerary rooms, while the second consists of one room, and the third consists of 6 rooms, while the largest one contains twenty rooms, These designs are related to the funerary rituals associated with each landmark according to the historical period in which each landmark was completed, so that, they date the construction period between the fifth and seventh centuries AD, the hypothetical reconstruction of the mentioned monuments was not very many, and the most important of them was done by the French architect Jean Claude Golvin, where he drawing the first funerary landmark (DJEDAR A) in Jabal al-Akhdar, which was a reconstruction of the general aspect in the method of hand drawing, where the engineer published His version of his work was less accurate and contained many errors, represented in the failure to include the entrance to the funerary chambers located in the upper pyramid part of the monument and without completing the general design of the monument, which appears from above to be a funerary complex, this work was praised by the researcher Jean-Pierre Laporte, as he stated in the book : Identités et Culture Dans l'Algérie Antique. that the graphic reconstruction carried out by the French architect is an amazing evocation and not a graphic reconstruction, after that The researcher Qadir also presented a new reconstruction and analyzed the location of the Djedar funerary monuments in Tiaret, but it was only a cope of what the French architect done, by presenting the small building opposite the entrance to the dome of the funerary chambers of monuments A , which was only a transfer and enlargement of what the architect Jean-Claude Golvain did with the addition of projections and cross-sections of the small building as an interior conception of the small building (Room G). that we see in the drawing of the architect Galvin, but it was criticized for its interpretations of the landmark and its architecture.

In this work, we will deal with a general presentation of the funerary monuments and their cultural&architectural identity, which some believe was built by the

Moors and others by the Roman Empire, and some of them inferred the Byzantine style of construction, which represents the design of the monuments, and then we will deal with a critique of the previous reconstruction works of djedar A, according to the current remains landmark and the architectural elements it highlights, that enable us to criticize what the previous researchers of the monument A have done, where the architectural identity of the monument is a key element in supporting the reconstruction argument that we will present using the aerial photogrammetry technique, the last of which we did in 2021 and the aim was to produce a digital model that will be raw material for future studies, as the digital model obtained will be a critical guide to the degree of reliability of the previous works of previous researchers, and we will also be able to analyze and deduce the architectural style, and use the ancient historical sources that mentioned the aforementioned monuments.

The latter will enable us to produce a reconstruction that has a high reliability rate, and we will also be able to provide a historical buildings information model for the monument A, and we are also trying to provide a reconstruction methodology for reconstruct archaeological monuments according to the reliability of the source, the latter of which can be used in deep training & learning and to support artificial intelligence tools that will generate archaeological reconstruction methodology's approaches with higher reliability.